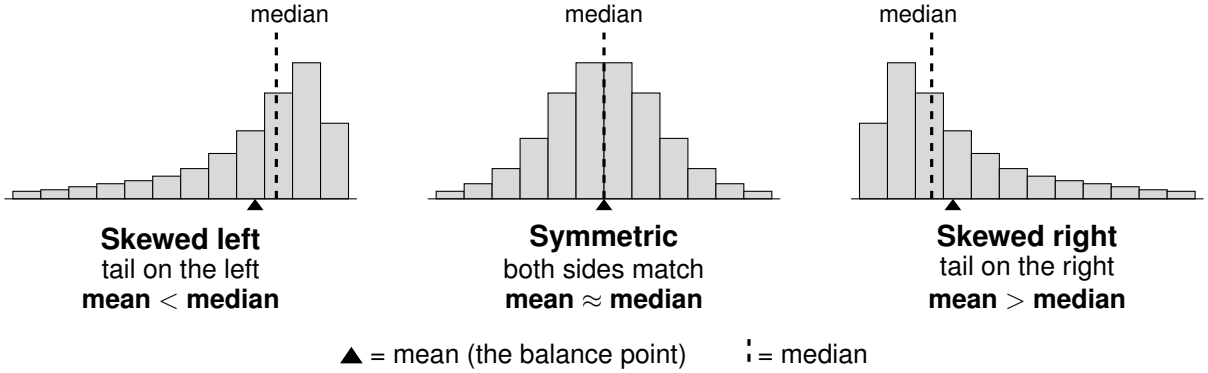
Mean, Median, and Shape KEY

Name _____

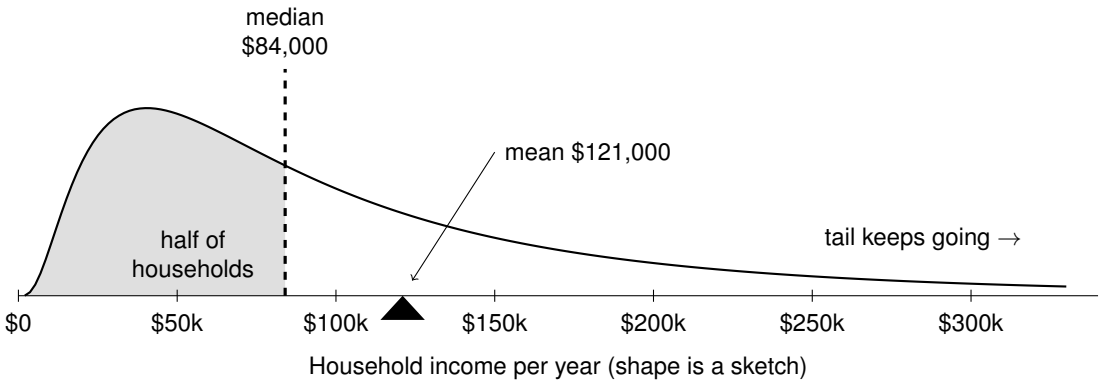
Mean (also called the **average**)	**Median** (the **middle**)
1. Add all the values. 2. Divide by how many values there are. Data: 2, 3, 8, 3, 4 $2+3+8+3+4=20$ $\quad 20 \div 5 = 4$ **Mean = 4** In symbols: $\bar{x} = \frac{\Sigma x}{n} = \frac{20}{5} = 4$	1. Put the values in order. 2. Find the middle value. In order: 2, 3, $\boxed{3}$, 4, 8 **Median = 3** Even number of values? Two numbers are in the middle. Find the number halfway between them. 2, $\boxed{3}$, $\boxed{5}$, 8 $\quad (3+5) \div 2 = 4$ **Median = 4**

Σ is the Greek capital letter **sigma**. It means **add up**. Σx means add up all the values. $\bar{x}$ ("x-bar") is the mean. n is how many values.

Shape changes the mean. The **tail** is the long, low side of a graph. The mean is **pulled toward the tail**. The median is not pulled much.



Real example: US household income (2024). Most households earn a medium amount. A small number of households earn millions of dollars. That makes a long tail on the right.



The median is a better description of a **typical** household. Most households earn **less than the mean**.

Source: US Census Bureau, *Income in the United States: 2024*. Median $83,730. Mean about $121,000.

Work with a partner. Compare answers when you finish each problem.

1. A student recorded how many hours she slept each night for one week.

   7, 8, 6, 9, 7, 5, 7

   (a) Find the mean. $49 \div 7 = 7$ hours

   (b) Find the median. 5, 6, 7, $\boxed{7}$, 7, 8, 9 $\rightarrow$ 7 hours

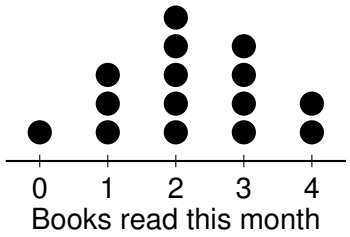
2. Six students timed their walk to the bus stop, in minutes.

   15, 22, 18, 30, 19, 16

   (a) Find the mean. $120 \div 6 = 20$ minutes

   (b) Find the median. 15, 16, $\boxed{18}$, $\boxed{19}$, 22, 30 $\rightarrow$ $(18 + 19) \div 2 = 18.5$ minutes. The 30 pulls the mean above the median.

3. This dot plot shows how many books some students read this month.

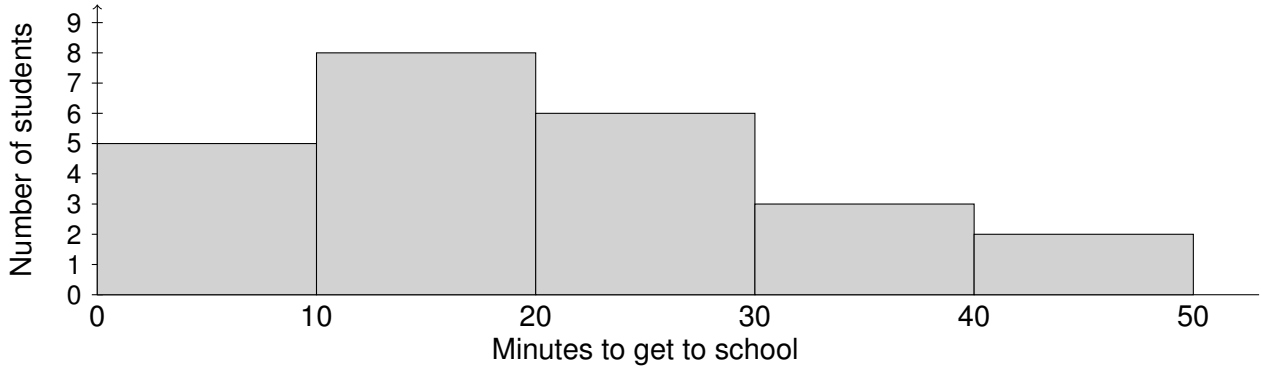


   (a) How many students are in the data? 15

   (b) Find the median. The 8th value is 2 books.

   (c) Find the mean. $(0 + 3 + 10 + 12 + 8) \div 15 = 33 \div 15 = 2.2$ books. Watch for: adding 0 to 4 instead of every dot.

4. This histogram shows how many minutes some students take to get to school.



Which bar holds the median? Explain.

10 to 20 minutes. There are $5 + 8 + 6 + 3 + 2 = 24$ values, so the median is between the 12th and 13th values. 5 values are below 10. $5 + 8 = 13$ values are below 20. So the 12th and 13th values are both in the 10 to 20 bar.

5. A small company has 5 people. Here is what each person is paid in one year.

Worker A	\$30,000
Worker B	\$32,000
Worker C	\$35,000
Worker D	\$38,000
Owner	\$265,000

(a) Find the mean pay. $400{,}000 \div 5 = \$80{,}000$

(b) Find the median pay. \$35,000

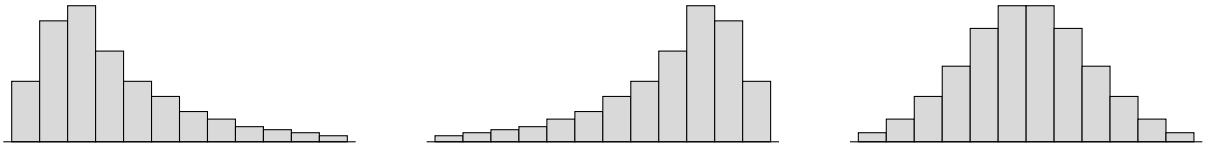
(c) The company puts this in a job ad: *"Average pay at our company: \$80,000!"* Is this true? Is it fair to a person looking for a job? Explain.

True, but not fair. 4 of the 5 people earn less than \$40,000. The owner's pay pulls the mean up. The median (\$35,000) is a better description of a typical worker.

(d) Leave out the owner. Find the mean pay of the 4 workers. How much did one person change the mean?

$135{,}000 \div 4 = \$33{,}750$. One person moved the mean up by \$46,250.

6. Look at each histogram. Name the shape. Then tell if the mean is greater than, less than, or about equal to the median.



A **B** **C**

Histogram	Shape	Mean $>$, $<$, or $\approx$ median?
A	skewed right	mean $>$ median
B	skewed left	mean $<$ median
C	symmetric	mean $\approx$ median

7. Guess the shape of each data set: skewed left, symmetric, or skewed right. Explain.

   (a) The price of every house sold in Columbia this year.
   Skewed right. Most houses are medium price. A few cost much more.

   (b) Scores on a very easy test. Most students score 90 to 100.
   Skewed left. Most scores are high. A few students score low.

   (c) The heights of all adult women in Missouri.
   Symmetric. Most heights are near the middle, with about as many tall as short.

8. Look at the US household income graph on page 1.

   (a) Why is the mean ($121,000) bigger than the median ($84,000)?
   Income is skewed right. A small number of very high incomes pull the mean toward the tail.

   (b) A news story says, *"The average US household earns $121,000."* Is this true? Does it describe a typical household? Explain.
   It is true for the mean. It does not describe a typical household. Most households earn less. The median is a better description of a typical household.

   (c) Do more than half of households earn less than $121,000? How do you know?
   Yes. Half of households earn less than $84,000 (the median). Some more earn between $84,000 and $121,000.